

Arbitrary-Field Minimal Rank for a Block Cycle: Literature Identification and an Independent Constructive Proof

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Abstract

Grossmann–Woerdeman ask whether the equality case in their block-cycle Theorem 3.2 remains true over an arbitrary field when the block size is $n - 1$ and the cycle holonomy has no fixed vector. A theorem of Cohen–Pereira, as explicitly summarized in Grossmann’s 2019 thesis, already supplies the required arbitrary-field upper bound; combined with the lower bound in the source, it answers the question. We also give a short independent construction. It proves the stronger formula that the normalized cycle has minimal rank $n - 1$ when its holonomy is the identity and n for every other invertible holonomy, over every field.

Status. `literature_implied_answer`, with an independent constructive proof. No novelty is claimed for the minimal-rank value itself.

1 Source question

Let $n \geq 2$, let $k = n - 1$, and consider the normalized partial block matrix over a field $\mathbb{F}$

$$\mathcal{A}_H = \begin{bmatrix} I & I & ? & ? & I & I & & \ddots & \ddots & \ddots \\ & & I & I & & & & & & \\ H & ? & \cdots & I & & & & & & \end{bmatrix}, \quad H \in GL_k(\mathbb{F}). \quad (1)$$

Thus the diagonal blocks and the blocks immediately above the diagonal are I_k , and the lower-left block is H .

Theorem 3.2 of Grossmann–Woerdeman [1] proves

$$\mathrm{mr}(\mathcal{A}_H) \geq \frac{nk - d}{n - 1}, \quad d = \dim \ker(H - I), \quad (2)$$

and obtains equality $\mathrm{mr}(\mathcal{A}_H) = n$ when $k = n - 1$, $d = 0$, and all eigenvalues of H lie in $\mathbb{F}$. Section 5.2, Problem 1, asks whether that conclusion holds over every field, without the eigenvalue-splitting hypothesis.

5.2. Open Problems. Here are some conjectures that remain unproven and questions that remain unanswered, which we find interesting.

- (1) Regarding theorem (3.2), we believe that the result should hold over any field $\mathbb{F}$, whether or not it is algebraically closed. In [4], we see that the result holds over any field when $n = 3$.
- (2) Our motivating conjecture from [2], which states that if $\mathcal{A}$ is a graph whose associated graph is bipartite chordal, then $\text{tmr}(\mathcal{A}) = \text{mr}(\mathcal{A})$, remains undecided. Of course, if $\text{tmr}(\mathcal{A}) = \text{mr}(\mathcal{A})$, then $\text{tmr}(\mathcal{A}) = \text{fmr}(\mathcal{A}) = \text{mr}(\mathcal{A})$.
- (3) It is notable that in our investigation of this particular patterns of partial matrices, we found that $\text{tmr}(\mathcal{A}) \neq \text{mr}(\mathcal{A})$ occurs if and only if $\text{tmr}(\mathcal{A}) < \text{fmr}(\mathcal{A}) < \text{mr}(\mathcal{A})$. We conjecture that this should be true in general. That

2 Why the question is already answered in the literature

Cohen–Pereira [2] study precisely a bipartite cycle with invertible specified blocks over an arbitrary base field and obtain a tight upper bound. The direct bridge is recorded by Grossmann himself in his 2019 thesis [3, pp. 64–65]: for the partial matrix in the source theorem, Cohen–Pereira give a completion whose rank is at most

$$\frac{nk - d}{n - 1}. \quad (3)$$

In the open-problem regime $k = n - 1$ and $d = 0$, this is the integer n . Together with (2), it gives

$$\text{mr}(\mathcal{A}_H) = n$$

over every field. Cohen–Pereira do not appear to frame their theorem as an answer to the later numbered question, so this packet records an implication rather than an explicit answer claim.

that $\text{fmr}(\mathcal{A}) \leq \frac{n}{n - 1}$. In particular, they show that if $\mathcal{A}$ is given as in (5.4) and the known blocks A_{ij} are invertible (which is certainly true if $A_{ij} = a_{ij}I$), then there exists a completion

of $\mathcal{A}$ whose rank is at most $\frac{nk - d}{n - 1}$.

3 Idea of the independent proof

A rank- n completion follows from factoring every completed block as $P_i Q_j$, where the Q_j are $(k + 1) \times k$ frames and each P_i is a common left inverse of two consecutive frames. The cycle condition turns this into a path from $[I; 0]$ to $[H^{-1}; 0]$ made of $k + 1$ transverse rank-one steps.

The path is constructed using one extra coordinate. First conjugate H^{-1} so that all its leading principal minors are nonzero. Lift the standard frame by a bottom row r . Then replace its columns one at a time by the target columns. Choosing the entries of r backwards makes each replacement transverse: at stage j the relevant determinant is affine in r_j , with nonzero coefficient equal to the j th leading principal minor.

4 The constructive frame lemmas

Lemma 1 (Strong leading-minor basis). *For every $T \in GL_k(\mathbb{F})$ there is an ordered basis in which every leading principal minor of the matrix of T is nonzero.*

Proof. We induct on k . Choose $v \neq 0$. There is a functional f with $f(v) = 1$ and $f(Tv) \neq 0$: if v, Tv are independent, prescribe the two values to be 1; if $Tv = \lambda v$, any $f(v) = 1$ works because $\lambda \neq 0$. Put $W = \ker f$. In a basis beginning with v and followed by a basis of W , write

$$T = \begin{bmatrix} a & b \\ c & D \end{bmatrix}, \quad a = f(Tv) \neq 0.$$

The Schur complement $S = D - ca^{-1}b$ is invertible because $\det T = a \det S$. By induction, choose a basis of W in which all leading principal minors of S are nonzero. In the resulting basis of the whole space, the leading minor of order $j \geq 2$ of T is a times the leading minor of order $j - 1$ of S ; the first one is a . All are nonzero. $\square$

Call full-column-rank $(k + 1) \times k$ matrices X, Y *adjacent* when $\text{rank}(Y - X) = 1$ and $\text{rank}[X \ Y] = k + 1$.

Lemma 2 (Lift-and-replace path). *For every $M \in GL_k(\mathbb{F})$ there exist full-column-rank frames*

$$X_0 = [I_k; 0], X_1, \dots, X_{k+1} = [M; 0]$$

such that every consecutive pair is adjacent.

Proof. Adjacency is preserved by left and right multiplication by invertible matrices. Lemma 1 therefore lets us assume that every leading principal minor Δ_j of M is nonzero. Let $e_1, \dots, e_k$ be the standard basis. We shall choose a row $r = (r_1, \dots, r_k)$ and use

$$X_0 = \begin{bmatrix} I \\ 0 \end{bmatrix}, \quad X_1 = \begin{bmatrix} I \\ r \end{bmatrix}.$$

For $1 \leq j \leq k$, obtain X_{j+1} from X_j by replacing column j of X_j by $[Me_j; 0]$. Before that replacement the columns are

$$[Me_1; 0], \dots, [Me_{j-1}; 0], [e_j; r_j], \dots, [e_k; r_k].$$

The replacement is transverse precisely when the determinant of these k columns together with $[Me_j; 0]$ is nonzero. After reordering columns and expanding in the bottom row, this determinant is a linear expression in $r_j, \dots, r_k$ whose coefficient of r_j , up to sign, is

$$\det[Me_1 \ \cdots \ Me_j \ e_{j+1} \ \cdots \ e_k] = \Delta_j \neq 0.$$

Choose $r_k, r_{k-1}, \dots, r_1$ in that order. At each stage there is exactly one forbidden value for the new coordinate, so another value is available over any field. At $j = k$ this also forces $r_k \neq 0$, hence $r \neq 0$ and the first lift is a rank-one transverse step. Every later difference changes one column and has rank one; the nonzero determinant proves transversality and preserves full column rank. The final frame is $[M; 0]$.

If the preliminary similarity was $C = S^{-1}MS$, apply the construction to C and send every frame X to $\text{diag}(S, 1)XS^{-1}$. This fixes $[I; 0]$ and sends $[C; 0]$ to $[M; 0]$. $\square$

Lemma 3 (Common left inverse). *Adjacent frames X, Y have a common left inverse: there is a $k \times (k + 1)$ matrix P such that $PX = PY = I_k$.*

Proof. Write $Y - X = w\varphi$ with $w \neq 0$ and $\varphi \neq 0$. Adjacency says $w \notin \text{im } X$, so $[X \ w]$ is invertible. Define $P = [I_k \ 0][X \ w]^{-1}$. Then $PX = I_k$, $Pw = 0$, and hence $PY = I_k$. $\square$

5 Stronger arbitrary-field formula

Theorem 4. *For the normalized block cycle (1), over an arbitrary field,*

$$\text{mr}(\mathcal{A}_H) = \begin{cases} n-1, & H = I_{n-1}, \\ n, & H \neq I_{n-1}. \end{cases}$$

In particular, the source question has an affirmative answer.

Proof. Put $k = n - 1$ and apply Lemma 2 to $M = H^{-1}$. Thus

$$X_0 = [I; 0], X_1, \dots, X_n = [H^{-1}; 0].$$

Set $Q_j = X_{j-1}$ for $1 \leq j \leq n$. For $i < n$, Lemma 3 gives P_i with $P_i Q_i = P_i Q_{i+1} = I$. For the final edge, it gives P_n with

$$P_n Q_n = I, \quad P_n (Q_1 H^{-1}) = I,$$

so $P_n Q_1 = H$. Therefore the block matrix

$$C = (P_i Q_j)_{i,j=1}^n = \begin{bmatrix} P_1 \\ \vdots \\ P_n \end{bmatrix} \begin{bmatrix} Q_1 & \cdots & Q_n \end{bmatrix}$$

is a completion of $\mathcal{A}_H$ and has rank at most $k + 1 = n$.

If $H \neq I$, then $d = \dim \ker(H - I) \leq k - 1 = n - 2$. The source lower bound (2) is strictly larger than $n - 1$, so the integer rank of every completion is at least n . Hence it equals n . If $H = I$, filling every block by I_k gives rank $k = n - 1$, and the specified identity block gives the matching lower bound. $\square$

The source normalizes an arbitrary cycle with invertible specified blocks by invertible block row and column operations. Hence Theorem 4 also applies to the original formulation, with H equal to the source's ordered product of specified blocks.

6 Verification, scope, and review recommendation

The accompanying script performs two checks. First, a breadth-first search over the entire frame graph finds the optimal $k + 1$ distance for every fixed-point-free endpoint for $(k, |\mathbb{F}|) = (1, 3), (2, 2), (2, 3), (3, 2)$. Second, it implements Lemmas 1 and 2 and verifies the constructed path for every invertible 2×2 matrix over $\mathbb{F}_2$ and $\mathbb{F}_3$, and every invertible 3×3 matrix over $\mathbb{F}_2$ (168 endpoints). These computations are checks, not dependencies of the proof.

The literature classification rests on the exact upper-bound attribution in Grossmann's thesis and the matching scope of Cohen–Pereira. The original Cohen–Pereira article is closed access in the sources checked; a reviewer should compare its theorem statement directly before final archival. The independent proof does not depend on that article, but it uses the lower bound from Theorem 3.2 of the source. Recommended status: `literature_implied_answer`; `constructive proof likely valid`.

References

- [1] B. W. Grossmann and H. J. Woerdeman, *Fractional minimal rank*, Linear and Multilinear Algebra 69 (2021), 19–39; arXiv:1710.07343. See Theorem 3.2 and Section 5.2, Problem 1.
- [2] N. Cohen and E. Pereira, *The cyclic rank completion problem with regular blocks*, Linear and Multilinear Algebra 66 (2018), 861–868, doi:10.1080/03081087.2017.1330863.

- [3] B. W. Grossmann, *Rank in Matrix Analysis: On the Preservers of Maximally Entangled States and Fractional Minimal Rank*, Ph.D. thesis, Drexel University, 2019, doi:10.17918/etd-9532.